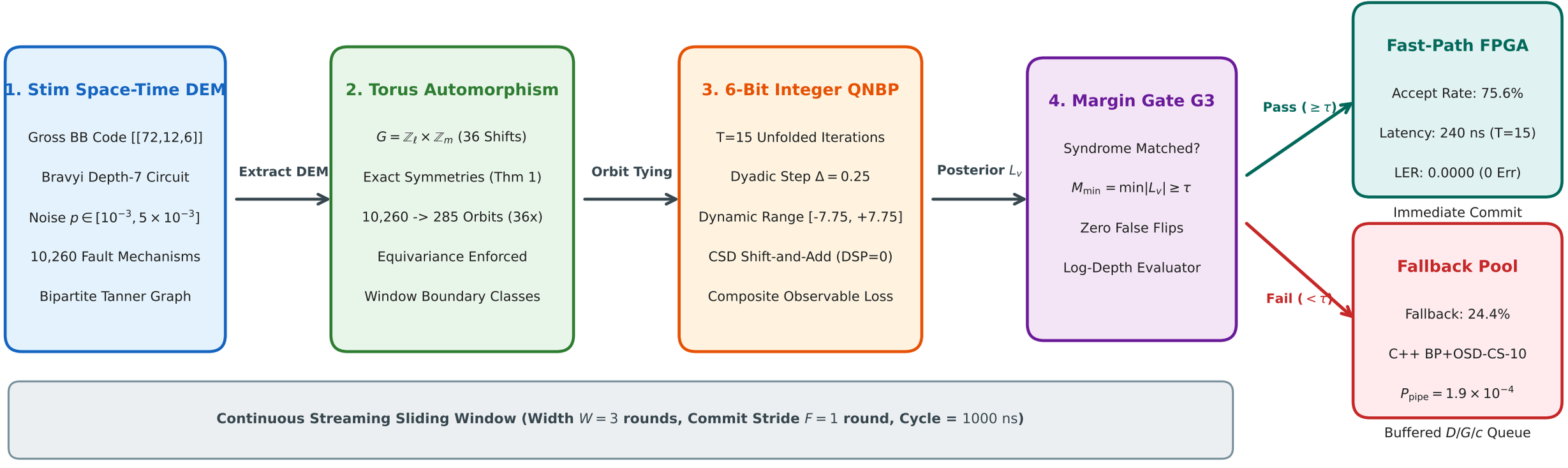
1. Stim Space-Time DEM
Gross BB Code [[72,12,6]]
Bravyi Depth-7 Circuit
Noise $p \in [10^{-3}, 5 \times 10^{-3}]$
10,260 Fault Mechanisms
Bipartite Tanner Graph
Extract DEM
2. Torus Automorphism
$G = \mathbb{Z}_\ell \times \mathbb{Z}_m$ (36 Shifts)
Exact Symmetries (Thm 1)
10,260 -> 285 Orbits (36x)
Equivariance Enforced
Window Boundary Classes
Orbit Tying
3. 6-Bit Integer QNBP
T=15 Unfolded Iterations
Dyadic Step $\Delta = 0.25$
Dynamic Range [-7.75, +7.75]
CSD Shift-and-Add (DSP=0)
Composite Observable Loss
Posterior L_v
4. Margin Gate G3
Syndrome Matched?
$M_{\min} = \min|L_v| \geq \tau$
Zero False Flips
Log-Depth Evaluator
Pass ($\geq \tau$)
Fast-Path FPGA
Accept Rate: 75.6%
Latency: 240 ns (T=15)
LER: 0.0000 (0 Err)
Immediate Commit
Fail ($< \tau$)
Fallback Pool
Fallback: 24.4%
C++ BP+OSD-CS-10
$P_{\text{pipe}} = 1.9 \times 10^{-4}$
Buffered $D/G/c$ Queue
Continuous Streaming Sliding Window (Width $W = 3$ rounds, Commit Stride $F = 1$ round, Cycle = 1000 ns)